Joshua Denbow | Milestone 4 | CS499 | 10/10/2026

Artifact Description

The artifact I selected is my InventoryApp, an Android application originally developed in my time taking CS 360: Mobile Architect & Programming. The application allows users to manage inventory by adding, editing, and deleting items while storing information in a local SQLite database. I've continued improving this application throughout my capstone, with this final enhancement focusing specifically on database design and functionality.

Justification and Improvements

I chose this artifact because it gave me a good opportunity to demonstrate how much my programming and database skills have improved since originally creating it. The original application had a fairly basic database structure, which worked but didn't offer much flexibility or functionality.

For this enhancement, I redesigned the SQLite database to use four related tables for users, categories, inventory, and inventory transactions. I added foreign keys, constraints, and transaction tracking to improve data organization and reliability. I also implemented SQL JOIN queries and reporting features that allow users to view low stock items, category summaries, and inventory history. These improvements made the application more useful while demonstrating my ability to work with relational databases.

Course Outcomes

This enhancement primarily demonstrates Course Outcome 4, which focuses on using appropriate computing techniques and tools to develop useful software solutions. By restructuring the database and implementing more advanced SQL operations, I was able to improve the application's functionality and reliability.

It also supports Course Outcome 3 through the design decisions involved in organizing the database and integrating it with the existing application. Course Outcome 5 is demonstrated through the use of database constraints, foreign keys, and transactions to help protect data integrity. I believe these improvements align well with my original enhancement plan, so I don't need to make any major changes to my planned outcome coverage.

Reflection on the Enhancement Process

Working on this enhancement helped me better understand how important database structure is to an application's overall design. One of the bigger challenges was updating the database without breaking the functionality I had already implemented in the previous enhancements. I had to make sure existing inventory information was preserved while introducing the new tables and relationships.

I also gained more experience using SQL queries, managing transactions, and connecting database operations to application features. After implementing the changes, I tested the application to confirm that the existing inventory, transaction history, reporting features, and restock recommendations were all working correctly. Overall, this enhancement helped me improve my understanding of relational database design and how it contributes to building more reliable and maintainable software.